

Agreeable but Rational: Benchmarking LLM Persuasion and Reasoning in Debating Competitions

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Abstract

We assess the agreeability and reasoning capabilities of large language models (LLMs) through a structured debating framework. Two LLMs engage in naturalistic, moderated debates across diverse topics, where one acts as persuader and the other as debater. While prior debate frameworks have studied specific vulnerabilities such as fallacy susceptibility, they have not been utilized as comprehensive benchmarks for reasoning capabilities, model comparison, or ideological entrenchment. We address this gap by introducing a benchmark that simultaneously probes reasoning quality, fallacy resistance, and ideological priors.

We find that LLMs exhibit high agreeability: when the same model acts as both persuader and debater, the debater switches positions in 62–100% of cases depending on configuration. To probe reasoning quality, we constrain the persuader to use either logical or fallacious arguments via helper agents. Contrary to expectations—and reversing findings from earlier studies—modern LLMs are increasingly resistant to fallacious reasoning: both helper types *decrease* persuasion success compared to unconstrained persuaders, with fallacious helpers achieving only 66% average success across all experiments versus 93% for unconstrained debate. We also observe a clear capability hierarchy in unconstrained debate, validating debate as a metric for relative reasoning strength: when a more advanced model persuades a weaker one, success reaches 99–100%, compared to **XX%** in reverse. Finally, we find that models exhibit “opinionated” behavior—fixed stances on certain claims that resist persuasion entirely. These opinionated patterns aggregate into significant ideological bias, indicating measurable training biases: liberal-leaning claims achieve 91% success versus 68% for conservative claims across models. Our framework establishes a benchmark for evaluating these dimensions in naturalistic adversarial settings.

Keywords: large language models; reasoning; persuasion; debate; logical fallacies; ideological bias; adversarial evaluation; benchmark

Introduction

Large language models (LLMs) have become indispensable tools for information retrieval, summarization, and open-ended dialogue. Their expanding use in high-stakes domains such as education, law, and decision-making has intensified scrutiny of their reasoning capabilities and behavioral reliability (Bang et al., 2024; Parmar et al., 2024). Early concerns centered on their tendency to agree with users too readily—a phenomenon termed *sycophancy* (Sharma et al., 2023)—and their vulnerability to fallacious or rhetorically manipulative inputs (Payandeh et al., 2023). Prior studies documented high stance-switching rates and frequent failures in logical rigor,

raising questions about the soundness of LLM-generated conclusions in adversarial contexts (Hong et al., 2025; Laban et al., 2023; Malmqvist, 2024).

Recent work has both reinforced and challenged these concerns. The LOGICOM framework (Payandeh et al., 2023) introduced a structured debate paradigm where two LLMs argue opposing positions on controversial claims, with one agent attempting to persuade the other. Using this approach, LOGICOM found that GPT-3.5 and GPT-4 were significantly more susceptible to fallacious arguments than logical ones. However, existing evaluation frameworks face fundamental limitations. Static benchmarks—such as multiple-choice reasoning tests or Q&A datasets (Parmar et al., 2024)—cannot probe how models reason under adversarial pressure or adapt their arguments dynamically. Single-dimension evaluations measure either logical competence or bias in isolation (Bang et al., 2024; Motoki et al., 2024), missing the complex interplay between reasoning quality, persuasion resistance, and ideological priors. Multi-agent debate has emerged as a promising paradigm for more naturalistic evaluation (Chan et al., 2024). We address this gap by using the structured debate paradigm as a unified benchmark that simultaneously evaluates (1) reasoning ability through argument quality and fallacy resistance, (2) opinionated behavior through asymmetric claim responses, and (3) ideological bias through differential success rates across political orientations. Our study extends the LOGICOM framework to newer models and introduces cross-model comparisons, enabling analysis of whether susceptibility to fallacies persists in modern LLMs and how capability differences affect persuasion dynamics.

We report several key findings. First, LLMs remain highly agreeable: when the same model acts as both persuader and debater, the responder changes its stance in 62–100% of cases depending on configuration, consistent with prior work on sycophantic behavior (Laban et al., 2023; Sharma et al., 2023). Second, persuasion ability varies markedly between models, with more advanced models persuading weaker ones at 99–100% success compared to **XX%** in reverse. Third, contrary to expectations, constraining the persuader to use either logical or fallacious strategies via helper agents *reduces* persuasion success compared to unconstrained persuasion, with fallacious helpers achieving only 66% success on average across all experiments versus 93% for unconstrained debate. This suggests modern LLMs are increasingly resistant to fallacious reasoning. Fourth, we find that models exhibit strong “opinionated” behaviors—59–71% of claims result in fixed outcomes regardless of arguments presented.

These opinions manifest as significant ideological bias: in same-model debates, liberal-leaning claims achieve 91% average success versus 68% for conservative claims, consistent with prior findings that LLMs exhibit left-leaning political tendencies (Motoki et al., 2024; Santurkar et al., 2023).

Taken together, our results challenge prevailing narratives about LLM brittleness (Malmqvist, 2024; Payandeh et al., 2023) and suggest that newer models are not only more logically competent but also less prone to manipulation than previously thought. More broadly, our framework demonstrates that structured debates provide a naturalistic, multi-dimensional benchmark: the same experimental setup reveals reasoning capabilities, fallacy resistance, capability asymmetries between models, and embedded ideological biases—dimensions that static evaluations cannot capture.

Background

Agreeability and Stance Switching in LLMs. Large language models (LLMs) often exhibit a high degree of agreeability, readily altering their stance in response to user prompts or minimal pressure. Studies using the *FlipFlop* benchmark show that models like GPT-4 and Claude 2 revise their answers in over 40% of cases when asked follow-up questions such as “Are you sure?” (Laban et al., 2023). These reversals are frequently accompanied by a decline in answer accuracy.

This pattern has been linked to social desirability bias. Research suggests LLMs adjust responses to align with traits deemed socially favorable, such as agreeableness and extroversion (Fanous et al., 2025; Salecha et al., 2024). Malmqvist (Malmqvist, 2024) notes that instruction tuning and RLHF may amplify this tendency, encouraging user-pleasing behavior over epistemic robustness. Recent work by Sharma et al. (Sharma et al., 2023) demonstrates that human preference data itself rewards sycophantic responses, explaining why RLHF-trained models exhibit this pattern. Hong et al. (Hong et al., 2025) further show that in multi-turn dialogues, alignment tuning amplifies stance-switching behavior.

Susceptibility to Fallacious Reasoning. While early LLMs were susceptible to various logical fallacies and reasoning errors, newer models have shown improved resistance.

The LOGICOM framework (Payandeh et al., 2023) specifically evaluated fallacy susceptibility through structured debates: a Persuader LLM attempts to convince a Debater LLM of a claim, with the Persuader either using logical or fallacious arguments. LOGICOM found that GPT-3.5 and GPT-4 were 41% and 69% more likely to be persuaded by fallacious arguments than logical ones, respectively. Our study uses the same debate paradigm but finds the opposite: fallacious reasoning now *harms* persuasion success. This striking reversal—using comparable methodology—suggests a clear evolution in LLM behavior since LOGICOM’s evaluation and highlights the need for updated benchmarks.

Logical Competence and Reasoning Ability. Recent instruction-tuned models have shown notable improvements in logical reasoning compared to their predecessors. These

improvements stem from both scaling and better alignment strategies. Benchmarks such as LogicBench (Parmar et al., 2024) systematically evaluate reasoning across diverse logical patterns, revealing that while models have improved, they still struggle with complex multi-step reasoning and negations.

Reasoning ability is increasingly robust in structured evaluations. Using the same debate paradigm as LOGICOM (Payandeh et al., 2023), we evaluate newer versions of similar-scale models (GPT-3.5-turbo and GPT-4.1-nano) to those in the original study. Our results show that these models are now significantly more resistant to fallacious thinking and perform best when reasoning is unconstrained. This reversal from LOGICOM’s findings marks a notable shift: while models are now more resistant to fallacious reasoning specifically, they remain highly agreeable overall, with 62–100% of same-model debates resulting in stance changes depending on configuration. This suggests improved discrimination between argument quality rather than increased resistance to persuasion generally.

Structured Debates and Adversarial Dialogue for Evaluation and Improvement. Structured debate frameworks offer a compelling method for benchmarking and improving LLM reasoning. Assigning models opposing roles surfaces differences in persuasive ability and reasoning strategies (Payandeh et al., 2023). Multi-agent debate has emerged as a promising evaluation paradigm; Chan et al. (Chan et al., 2024) demonstrate that LLM “referee teams” that debate text quality achieve higher correlation with human judgments than single-agent approaches.

Beyond evaluation, these setups enable training protocols that improve stability. Debate contexts also serve as diagnostic tools, helping expose overconfidence and inconsistency (Laban et al., 2023), and may inform future approaches to model evaluation and training in multi-turn adversarial settings.

Results

We evaluate LLM behavior in structured debates using the framework illustrated in Figure 1. Each debate involves a *Persuader* agent attempting to convince a *Debater* agent of a given claim. Debates proceed in rounds: the Persuader presents an argument, the Debater responds, and a Conviction Moderator evaluates the Debater’s conviction level on a 1–10 scale. This exchange continues until the Debater is convinced or a maximum of 10 rounds is reached.

Three moderator agents oversee each debate. A *Topic Moderator* (GPT-4.1-mini) monitors whether the discussion stays on-topic and determines if the debate should continue. A *Conviction Moderator* (GPT-4.1-mini) evaluates whether the Debater has been convinced and assigns a conviction score (1–10). A *Quality Moderator* (o4-mini) rates argument quality per round and provides an overall debate quality assessment at the end. The Persuader may optionally receive suggestions from a *Helper* agent, configured to suggest logical arguments, fallacious tactics, or provide no assistance.

We tested eight model configurations: GPT-4.1, GPT-4.1-

Fallacy / Logical / None

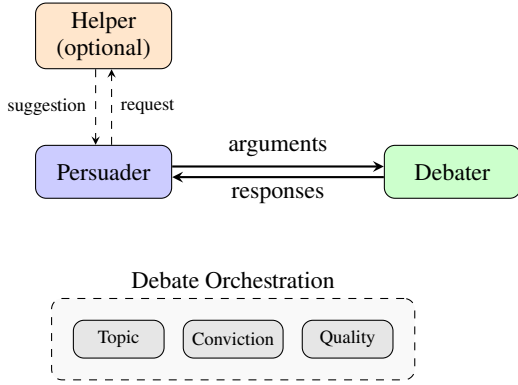


Figure 1: Debating framework architecture. Each round: (1) the Persuader optionally requests suggestions from a Helper agent, (2) the Persuader sends arguments to the Debater, (3) the Debater responds. The Debate Orchestration layer manages topic relevance, conviction assessment, and argument quality evaluation.

nano, and GPT-3.5-turbo in pairwise combinations as Persuader and Debater. For each configuration, we ran debates across 200 claims with all three helper conditions, yielding 4,800 total debates. This design allows us to measure not only overall persuasion success but also capability asymmetries between models and differential responses to argument types.

Stance Switching Behavior

LLMs exhibit high agreeability across all configurations: in same-model debates, between 62% and 100% of debates end with the Debater changing their initial stance, depending on helper type (Table 1). This confirms that modern LLMs remain highly persuadable in adversarial dialogue contexts, consistent with prior findings on sycophantic behavior (Laban et al., 2023; Sharma et al., 2023).

Dichotomous Switching Patterns. We observe a dichotomous pattern in stance switching: in many debates, the Debater either agrees immediately in round 1 or maintains their position through all 10 rounds, with relatively few debates ending in intermediate rounds (Figure 2). This pattern is particularly pronounced in more advanced models—GPT-4.1 and GPT-4.1-nano show strong bimodal distributions with peaks at rounds 1 and 10, while GPT-3.5-turbo exhibits a more gradual distribution across rounds. This suggests that newer models may have stronger prior dispositions toward certain claims: when predisposed to agree, persuasion succeeds instantly; when predisposed to disagree, the model resists throughout. This behavior requires further investigation, but a potential explanation is that more capable models have heightened biases toward specific opinions—a hypothesis we examine below in the context of ideological claims.

Helper Agents Decrease Persuasion Success

Across all model configurations, unconstrained persuaders (*no helper*) consistently outperformed both helper-assisted conditions (Table 1). This pattern holds regardless of whether the helper suggests logical or fallacious arguments.

Configuration	No Helper	Logical	Fallacy
GPT-4.1 → GPT-4.1	99.5%	95.5%	62.0%
GPT-4.1 → GPT-4.1-nano	99.5%	97.5%	66.0%
GPT-4.1 → GPT-3.5-turbo	100.0%	100.0%	52.0%
GPT-4.1-nano → GPT-4.1	94.5%	96.4%	78.0%
GPT-4.1-nano → GPT-4.1-nano	83.3%	83.0%	75.0%
GPT-4.1-nano → GPT-3.5-turbo	99.0%	93.9%	67.0%
GPT-3.5-turbo → GPT-4.1-nano	76.5%	66.0%	62.8%
GPT-3.5-turbo → GPT-3.5-turbo	90.0%	71.7%	68.0%
Average	92.8%	88.1%	66.2%

Table 1: Persuasion success rates by helper type and model configuration. No Helper consistently outperforms helper-assisted conditions across all eight experiments.

The No Helper advantage ranges from 8 to 32 percentage points over Fallacy Helper. This suggests that constraining persuasion strategies—whether logical or fallacious—limits the natural flexibility that makes unconstrained persuasion more effective.

Fallacy Ineffectiveness

Fallacy Helper not only produces the lowest success rates but also the lowest debate quality ratings. Average debate quality for Fallacy Helper was 5.6/10 across all experiments, compared to 7.0/10 for No Helper.

Analysis of specific fallacy types reveals differential resistance (Table 2). “False Dilemma” and “Causal Fallacy” achieve the highest success rates (54–55%), while most other fallacy types cluster between 35–43%. “Ad Hominem” is least effective (35.8%), suggesting that direct personal attacks are particularly ineffective against modern LLMs. The overall pattern indicates that while models show some vulnerability to binary framing (False Dilemma), they are broadly resistant to manipulative tactics.

Fallacy Type	Usage %	Success %
False Dilemma	22.0	54.6
Appeal To Popular Opinion	15.6	43.1
Appeal To Emotion	15.5	40.9
Appeal To Authority	14.6	41.0
False Information	10.4	39.1
Ad Hominem	10.2	35.8
Straw Man	6.4	37.2
Slippery Slope	4.4	38.3
Causal Fallacy	1.1	54.3

Table 2: Per-fallacy usage and success rates across all experiments. False Dilemma is both the most frequently used (22%) and most effective (54.6%). Ad Hominem, despite being commonly used (10.2%), is least effective (35.8%).

Distribution of Rounds to Completion

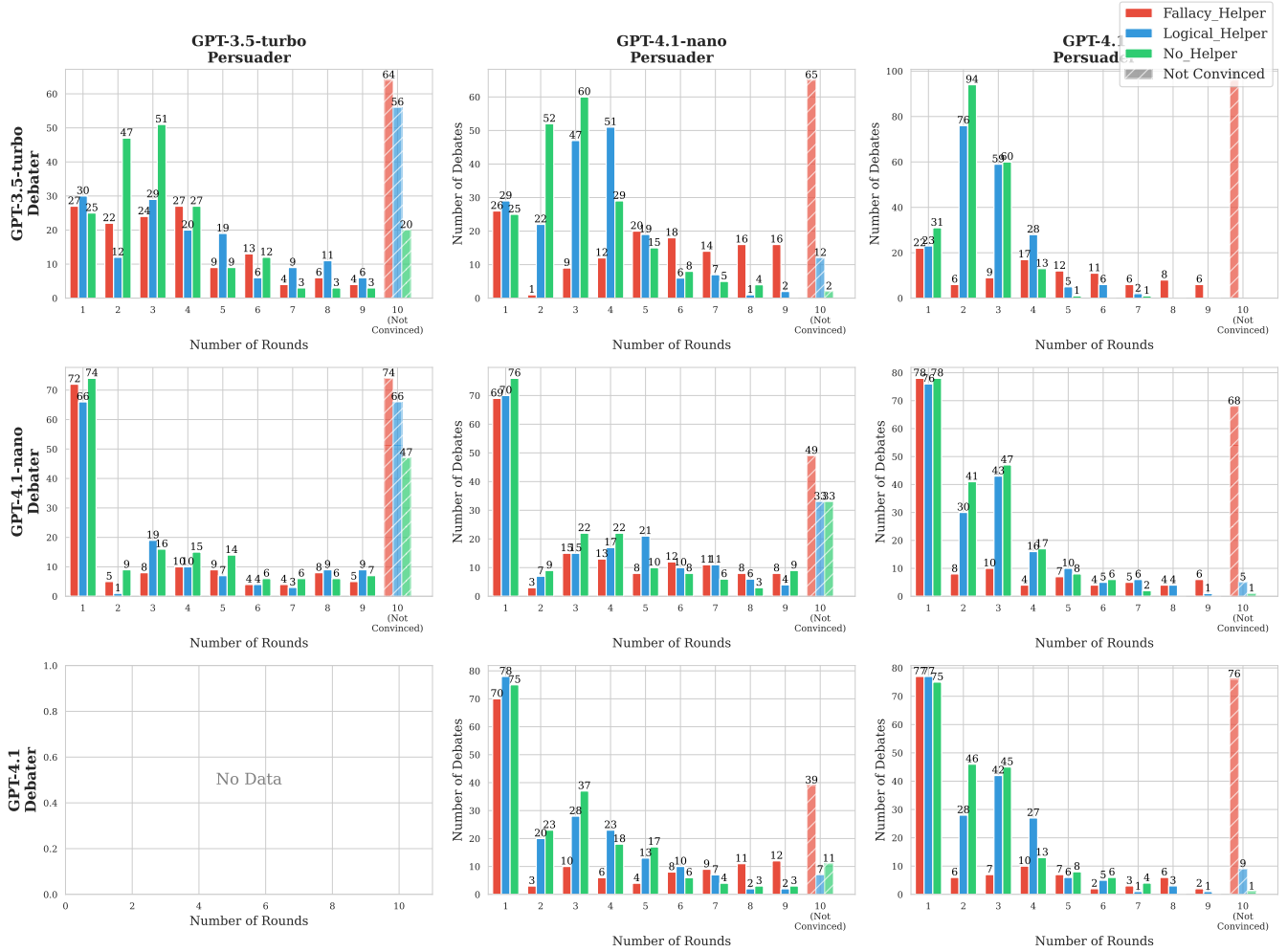


Figure 2: Distribution of rounds to debate completion across model configurations. More advanced models (GPT-4.1, GPT-4.1-nano) show pronounced bimodal distributions with peaks at round 1 (immediate agreement) and round 10 (maximum resistance), while GPT-3.5-turbo exhibits more gradual conviction patterns.

Model Capability Asymmetry

In unconstrained debate, when a more capable model persuades a less capable one, success rates reach 99–100%. Conversely, when the less capable model attempts to persuade the more capable one, success drops to **XX%** (Figure 3). This asymmetry suggests our framework can serve as a benchmark for relative model capability in adversarial reasoning contexts.

Opinionated Behavior and Ideological Bias

Beyond general agreeability, we observe that models exhibit “opinionated” behavior—holding fixed stances on certain claims that resist persuasion entirely. Our 200 claims are organized as 100 opposing pairs (e.g., “X should be mandatory” vs. “X should not be mandatory”), allowing us to identify asymmetric responses where models readily accept one position while resisting its opposite.

Asymmetric Claim Pairs. We find 7 claim pairs with 100% asymmetry: one claim succeeds in every debate while its opposite fails in every debate. An additional 29 pairs show strong asymmetry ($\geq 60\%$ difference in success rates). This pattern holds even when stronger models attempt to convince weaker ones, leading us to believe that the asymmetry is a result of the debater’s pre-existing dispositions rather than persuader capability.

Aggregate Ideological Bias. These asymmetries aggregate into significant ideological bias. Liberal-leaning claims achieve substantially higher success rates than conservative-leaning claims across all models (Table 3). This bias persists across all experimental configurations, including cross-model debates where more capable persuaders attempt to convince less capable debaters.

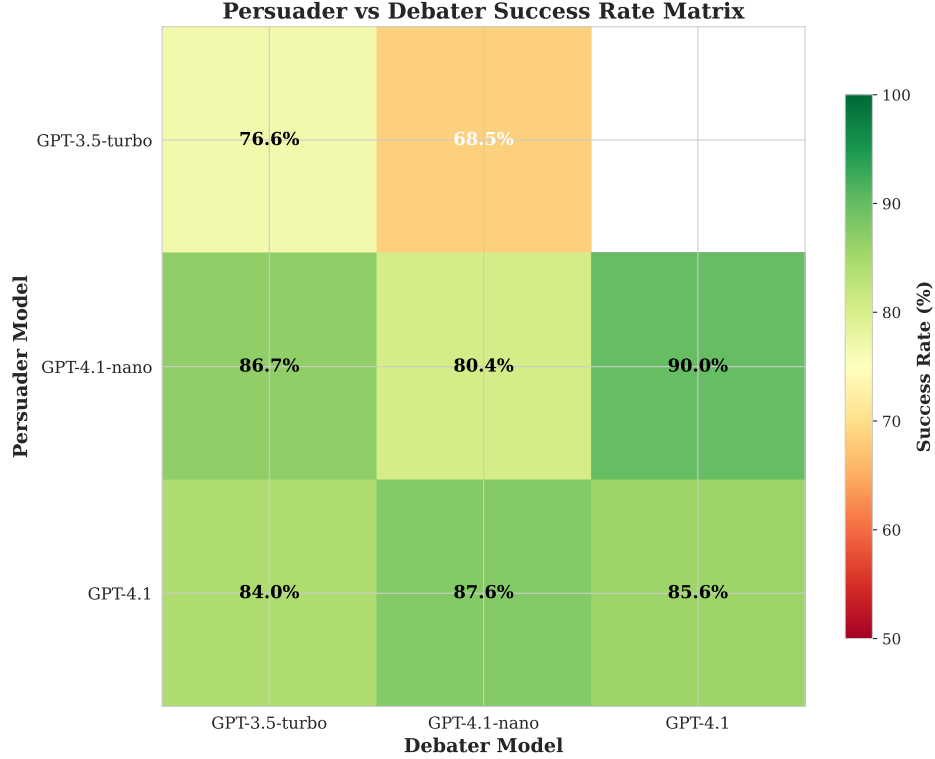


Figure 3: Model capability matrix showing overall persuasion success rates for each Persuader (row) vs. Debater (column) configuration across GPT-3.5-turbo, GPT-4.1-nano, and GPT-4.1. Diagonal entries show self-debate; off-diagonal entries reveal capability asymmetry—stronger models (bottom-right) consistently outperform weaker ones (top-left).

Model (self-play)	Liberal	Conservative	Neutral
GPT-4.1	95.7%	69.9%	90.3%
GPT-4.1-nano	88.0%	66.7%	85.0%
GPT-3.5-turbo	87.6%	66.1%	76.2%

Table 3: Ideological bias in self-play persuasion success rates by claim ideology. All models show a consistent liberal advantage of 20–26 percentage points over conservative claims ($p < 0.0001$ for all comparisons).

Figure 4 visualizes this bias across model configurations. These findings align with prior research documenting liberal-leaning political bias in LLMs. Motoki et al. (Motoki et al., 2024) found systematic left-leaning bias in ChatGPT across multiple countries, while Santurkar et al. (Santurkar et al., 2023) demonstrated that RLHF training introduces left-leaning political views. Fulay et al. (Fulay et al., 2024) further showed that training for truthfulness on standard datasets inadvertently amplifies liberal bias. The combination of asymmetric claim pairs and aggregate success rate differences suggests that ideological biases are deeply embedded in model behavior and persist regardless of the persuader’s capability.

Discussion

Our findings present a striking departure from prior work on LLM reasoning and persuasion. Using the same structured

debate paradigm as LOGICOM—where a Persuader LLM attempts to convince a Debater LLM of a claim—we find directly opposite results. LOGICOM (Payandeh et al., 2023) reported that fallacious arguments were 41–69% *more* effective at persuading GPT-3.5 and GPT-4 than logical ones. Our results show the reverse: fallacious helper agents consistently produce the *lowest* success rates across all configurations. This reversal, observed using comparable methodology, likely reflects improvements in model alignment and RLHF training that have increased resistance to fallacious reasoning patterns in the intervening period.

Viewed through the lens of dual-process theory (Kahneman, 2011), our results suggest that modern LLMs may be engaging in something analogous to System 2 (deliberative) processing rather than System 1 (intuitive) responses. Fallacies typically exploit heuristic shortcuts characteristic of System 1; resistance to such tactics implies careful argument evaluation. Similarly, the Elaboration Likelihood Model (Petty & Cacioppo, 1986) predicts that central route processing (careful scrutiny of arguments) should favor logical over fallacious content. The pattern we observe—with False Dilemma achieving the highest success (54.6%) and Ad Hominem the lowest (35.8%)—suggests models may be more susceptible to binary framing than to direct personal attacks, consistent with training objectives that emphasize respectful discourse.

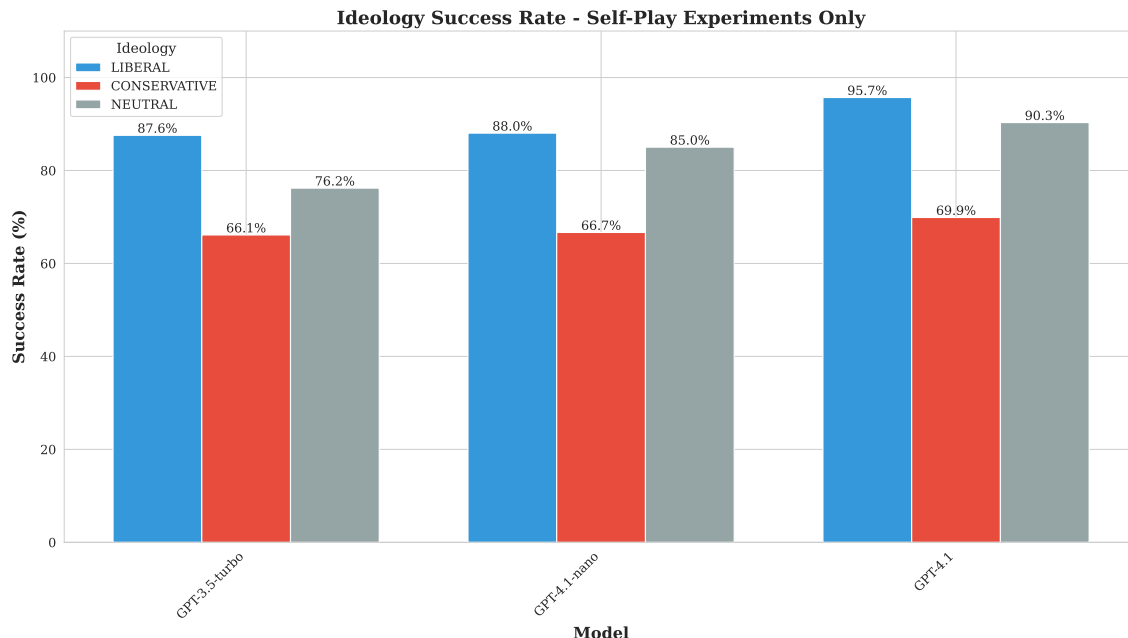


Figure 4: Ideological bias in self-play experiments (same model as persuader and debater). Liberal claims consistently achieve higher success rates than conservative claims across all three models.

An alternative explanation draws on recent mechanistic interpretability research suggesting that LLMs reduce complex arguments to low-dimensional representations. Marks and Tegmark (Marks & Tegmark, 2023) discovered that models project inputs onto a linear “truth direction” in activation space, effectively reducing multi-faceted debates to a single axis. Similarly, Sarfati et al. (Sarfati et al., 2025) show that reasoning in LLMs unfolds along low-dimensional trajectories (~ 10 dimensions) rather than branching logical trees—the model navigates a geometric manifold rather than performing symbolic inference. This “flattened” reasoning may explain both fallacy resistance and sycophancy: Belinkov (Belinkov et al., 2025) demonstrates that models maintain competing “truth” and “sycophancy” directions, with outputs determined by which axis dominates during generation. Under this view, models resist manipulation not through genuine argument evaluation, but because fallacious content projects poorly onto their learned truth representations. This interpretation is consistent with our finding that models remain highly agreeable overall: they avoid manipulation not by detecting it, but by defaulting to responses that align with their internal “safe” directions when faced with potentially problematic arguments.

The finding that *both* helper types underperform compared to unconstrained persuasion is perhaps more surprising. We hypothesize that helper constraints—whether logical or fallacious—over-specify the persuasion strategy, limiting the Persuader’s adaptive flexibility. The unconstrained Persuader can dynamically adjust tactics based on Debater responses, while helper-constrained Persuaders are locked into

prescribed approaches.

The capability asymmetry between model configurations (99–100% when a more advanced model persuades a weaker one vs. **XX**% in the reverse) demonstrates that our framework can distinguish model capabilities in adversarial reasoning. This suggests potential applications in model evaluation and capability benchmarking beyond traditional static assessments.

The significant ideological bias observed in both models raises important concerns for deployment in sensitive domains. Liberal claims succeeded at substantially higher rates than conservative claims, with the bias more pronounced in the more recent model. This pattern is consistent with broader findings that LLMs exhibit measurable political bias in both content and style (Bang et al., 2024). Whether this reflects training data composition, RLHF reward patterns, or other factors warrants further investigation.

Limitations. Our current experiments use only three OpenAI models; extending to other model families (Claude, Gemini, open-weight models) would strengthen generalizability. The ideological labels, while produced through a multi-agent consensus process, may still reflect labeling biases inherent to the models used for annotation.

Methods

Debating Framework Architecture

Our framework orchestrates debates between two LLM agents: a *Persuader* attempting to convince a *Debater* of a given claim. Each debate proceeds in rounds, with the Persuader presenting arguments and the Debater responding.

Three *Moderator* agents oversee the debate: a *Topic Moderator* that ensures arguments remain on-topic and determines when to continue the debate, a *Conviction Moderator* that evaluates the Debater’s conviction level (1–10 scale) after each round, and a *Quality Moderator* that rates argument quality per round and overall debate quality. Debates terminate when the Debater is convinced, reaches a maximum of 10 rounds, or goes off-topic.

Helper Agents

We test three Persuader configurations: (1) *No Helper*, where the Persuader argues freely; (2) *Logical Helper*, which suggests logically structured arguments; and (3) *Fallacy Helper*, which suggests fallacious tactics including ad hominem attacks, appeals to emotion, false information, causal fallacies, slippery slopes, appeals to authority, appeals to popular opinion, straw man arguments, and false dilemmas.

Model Configurations

We pit models against each other in nine configurations using GPT-3.5-turbo, GPT-4.1-nano, and GPT-4.1 as both Persuader and Debater in all pairwise combinations (including self-play). This design allows us to measure capability asymmetries: when a more capable model persuades a less capable one (and vice versa).

Claim Dataset and Ideological Tagging

We use the same 200 claims used by LOGICOM (Payandeh et al., 2023), originally drawn from Habernal et al.’s dataset of debates from The New York Times (Habernal et al., 2018), organized as 100 opposing pairs (e.g., “X should be mandatory” vs “X should not be mandatory”). To label ideological leaning, we employed a multi-agent consensus approach: three LLM agents (Claude-4.5-opus+gemini-3-pro+GPT-5.2) independently rated each claim on a 1–5 scale from conservative to liberal with written justifications. A fourth agent (Claude-4.5-opus) reviewed these ratings to produce a final label + confidence score, accounting for potential tagging bias. We ensured opposing claims received opposite ideological labels. The majority of labels (75%) were assigned with high confidence (Figure 5). The final distribution: 62 Liberal, 62 Conservative, and 76 Neutral claims.

Acknowledgments

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Author contributions

G.S. and J.I. designed and implemented the framework, conducted experiments, and analyzed the data. S.R. supervised the project and contributed to experimental design. All authors contributed to writing the manuscript.

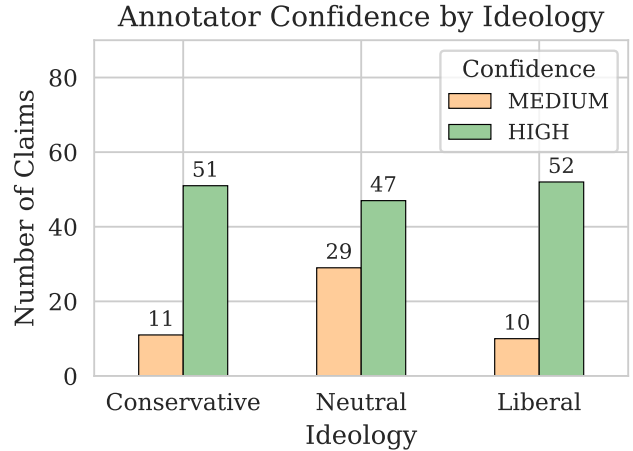


Figure 5: Annotator confidence distribution by ideology label. The majority of claims (75%) were labeled with high confidence across all ideology categories.

Competing interests

The authors declare that they have no competing interests.

Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the paper. Code and data will be made available upon publication.

References

- Bang, Y., Chen, D., Lee, N., & Fung, P. (2024). Measuring political bias in large language models: What is said and how it is said. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 10846–10867. <https://doi.org/10.48550/arXiv.2403.18932>
- Belinkov, Y., et al. (2025). LLMs know more than they show: On the intrinsic representation of LLM hallucinations. *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=b0b1cfc8ede53f452cabf8b9cf4eef76>
- Chan, C.-M., Chen, W., Su, Y., Yu, J., Xue, W., Zhang, S., Fu, J., & Liu, Z. (2024). ChatEval: Towards better LLM-based evaluators through multi-agent debate. *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=FQepisCUWu>
- Fanous, J., et al. (2025). Social sycophancy: A broader understanding of llm sycophancy. *arXiv preprint arXiv:2505.13995*. <https://doi.org/10.48550/arXiv.2505.13995>

- Fulay, S., Brannon, W., Ozdoganlar, O. B., Mathew, L., Kabbara, J., Binnquist, T., Tsouvalas, V., McIlroy-Young, R., Faltings, B., Anderson, A., & Roy, D. (2024). On the relationship between truth and political bias in language models. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 8954–8974.
- Habernal, I., Wachsmuth, H., Gurevych, I., & Stein, B. (2018). The argument reasoning comprehension task: Identification and reconstruction of implicit warrants. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 1930–1940. <https://doi.org/10.18653/v1/N18-1175>
- Hong, J., Byun, G., Kim, S., & Shu, K. (2025). Measuring sycophancy of language models in multi-turn dialogues. *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2239–2259. <https://doi.org/10.18653/v1/2025.findings-emnlp.121>
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus; Giroux.
- Laban, P., Murakhov'ska, L., Xiong, C., & Wu, C.-S. (2023). Are you sure? challenging llms leads to performance drops in the flipflop experiment. *arXiv preprint arXiv:2311.08596*. <https://doi.org/10.48550/arXiv.2311.08596>
- Malmqvist, L. (2024). Sycophancy in large language models: Causes and mitigations. *arXiv preprint arXiv:2411.15287*. <https://doi.org/10.48550/arXiv.2411.15287>
- Marks, S., & Tegmark, M. (2023). The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*. <https://doi.org/10.48550/arXiv.2310.06824>
- Motoki, F., Pinho Neto, V., & Rodrigues, V. (2024). More human than human: Measuring ChatGPT political bias. *Public Choice*, 198, 3–23. <https://doi.org/10.1007/s11127-023-01097-2>
- Parmar, M., Patel, N., Varshney, N., Nakamura, M., Luo, M., Mashetty, S., Mitra, A., & Baral, C. (2024). LogicBench: Towards systematic evaluation of logical reasoning ability of large language models. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 13679–13707. <https://doi.org/10.18653/v1/2024.acl-long.739>
- Payandeh, M., Gao, X., Zhao, Y., McKenzie, P., Pavlick, E., Roth, D., & Linzen, T. (2023). Logicom: Evaluating llms on interactive logical reasoning and fallacies. *arXiv preprint arXiv:2308.09853*. <https://doi.org/10.48550/arXiv.2308.09853>
- Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In *Advances in experimental social psychology* (pp. 123–205, Vol. 19). Academic Press.
- Salecha, A., Ireland, M. E., Subrahmanya, S., Sedoc, J., Ungar, L. H., & Eichstaedt, J. C. (2024). Large language models show human-like social desirability biases in survey responses. *arXiv preprint arXiv:2405.06058*. <https://doi.org/10.48550/arXiv.2405.06058>
- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? *Proceedings of the 40th International Conference on Machine Learning*, 202.
- Sarfati, R., Liu, T. J. B., Boullé, N., & Earls, C. J. (2025). Lines of thought in large language models. *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=649f080d8891ab4d4b262cb9cd52e69a>
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askill, A., Bowman, S. R., Cheng, N., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., & Perez, E. (2023). Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*. <https://doi.org/10.48550/arXiv.2310.13548>